Document Information	
Filename	TP_F100U-00048_User_Guide_gPTP_Feature_User_Guide
Status	Release
Revision Date	July 1, 2020
Version	01.00
Document ID	N/A

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1 Introduction

This document provides usage information for (g)PTP features of Ethernet MAC as described in IEEE802.1AS (Timing and Synchronization for Time-Sensitive Applications in Bridget Local Area Networks). It is a profile of IEEE1588. A profile defines “The set of allowed Precision Time Protocol (PTP) features applicable to a device”.

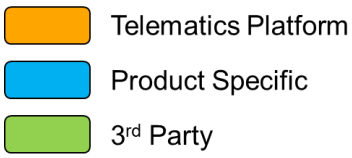
The target audience for this document are to use g(PTP) features of Ethernet MAC. A test setup for PTP time synchronization between master and slave is also explained .

1.1 Notational Conventions

Ctrl+x	*A sequence with a ‘+’ indicates that you must press and hold one key or pointing device button while pressing another key or pointing device button*
[ ]	*Optional elements in a command*
…..	*Indicates one of the following possibilities:* • *Additional optional arguments in a statement have been omitted* • *The preceding item or items can be repeated one or more times* • *Additional parameters, values, or other information can be entered*
-	*Line continuation: indicates that a command line or code line continues on the following line when at the end of a line*
Italic text	*Indicates important information, complete titles of manuals or variables*
`Mono-spaced`	*Example code or interactive screen display/output*
Numbers	*All numbers are assumed to be decimal unless otherwise noted. Non decimal radixes – binary, octal or hexadecimal – are explicitly indicated*
Underline blue text	*A link to an external reference (e.g., website, document)*

`Modules`
`|-> section`
`|-> item`
	A link to a section or item in the Telematics Platform html interface specification (Doxygen output from platform code headers)

Diagram Legend



2 Setup

2.1 Tools/Equipment Needed

Tools/Equipments for the test setup explained in the document

- Two Continental targets that could be flashed, with necessary harness / test fixture.
- 12-volt power supply
- High speed USB cable for access to Android Debug bridge of NAD module
- USB cable to connect to serial port of NAD module
- Personal Computer (PC) with Windows 7 or Windows 10 operating system, note that Windows 10 is recommended (preferred)
- ADB (Refer Documentation here)

Note – The test setup explained here uses a Continental target module as master device. This could be replaced by any other ECU to be gPTP master to test behavior of Continental gPTP slave module.

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2.2 Setup Instructions

Setup instructions for the test setup explained in the document

1. Install USB host driver compatible with ADB
2. Install Android Debug Bridge (ADB) on PC.
3. Connect the HS USB port of NAD module to be configured as PTP master to the PC via High Speed USB cable.
4. Execute adb shell on windows command prompt and enter into command shell of sa515 NAD target to be configured as PTP master.
5. Connect USB Serial port of NAD module to be configured as PTP slave to the PC and access using Putty.

2.3 Setup Troubleshooting

Not Applicable

3 Features/Components

3.1 (g)PTP

3.1.1 Overview

3.1.1.1 Platform Content Overview

The platform Ethernet driver interprets the PTP protocol for upper layer services. The Ethernet driver while acting as gPTP slave, the timestamps received form gPTP master and synchronization count are passed onto the upper layer. Platform provides gPTP daemon of open AVB stack in the upper layer for using gPTP features.

3.1.1.2 Product Expectations

- Configure and bring up Ethernet interface for PTP master/slave.
- Connect master and slave through Ethernet interface

3.1.1.3 Extending Platform Functionality

Platform software has integrated Open AVB Stack from AVNU Alliance. It uses the gptp daemon.

Reference documentation and websites

- Open AVNU Website
- (g)PTP helper API
- Sample test example for (g)PTP API

3.1.2 Setup

3.1.2.1 Tools/Equipment Needed

Not Applicable

3.1.2.2 Setup Instructions

Not Applicable

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3.1.2.3 Setup Troubleshooting

Not Applicable

3.1.3 Usage

3.1.3.1 Basic Instructions

To test gPTP daemon, a PTP master is connected to a PTP slave through Ethernet interface. Two NAD modules could be used as PTP master and PTP slave for the test setup. Please refer Setup

3.1.3.2 Detailed Instructions for Master NAD module to test gPTP daemon

The gPTP daemon (daemon_cl) uses syslog for logging. So below command could be used to view the daemon_cl logs on system console.

```
[root@sa415m system]#tail -f /data/log/messages | grep daemon_cl &
```

Daemon_cl could be run with different options. The command below uses

-E – daemon_cl executes in test mode

-R 1 – Assigns priority as 1 and make the device a gPTP master

```
[root@sa415m system]#daemon_cl eth0 -R 1 -E
```

The logs of gPTP master are in embedded file below.

Logs-gPTPMaster.txt

Note – Master gPTP device do not synchronize time and hence Sync Count is 0 as in the below log snippet..

Master Device #

2020-06-03 03.57.09.154 daemon_cl[][info] STATUS : GPTP [03:57:09:153] Clock offset:0 Clock rate ratio:1.000000 Sync Count:0 PDelay Count:33
2020-06-03 03.57.09.154 daemon_cl[][info] STATUS : GPTP [03:57:09:153] MASTER Clock offset:0 Clock rate ratio:1.000000 **Sync Count:0** PDelay Count:33

3.1.3.3 Detailed instructions for slave NAD module to test gPTP daemon

```
[root@sa515m ~]# tail -f /data/log/messages | grep daemon_cl &
[root@sa515m ~]# daemon_cl eth0 -R 255 -E
```

-R 255 – Assigns priority 255 and makes the device a gPTP slave

Logs-gPTPSlave.txt

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Note – Master clock offset is 18 at syn count 47 in the below log snippet of gPTP slave device. The offset becomes 0 by sync count 49.

Slave Device #

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:116] MASTER Clock offset:18 Clock rate ratio:1.000000 Sync Count:47 PDelay Count:7

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:120] SYSTEM Clock offset:-2833129131207299 (avg:-2772849788286335) Clock rate ratio:0.999896 avg(0.999900) Sync Count:47 PDelay Count:7

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:121] QTIMER Clock offset:-1591146636257002424 (avg:-1591086356913385231) Clock rate ratio:0.999896 avg(0.999900) Sync Count:47 PDelay Count:7

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:125] Adjust clock rate ppm:-4.314614

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:247] Clock offset:1 Clock rate ratio:1.000000 Sync Count:48 PDelay Count:7

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:247] MASTER Clock offset:1 Clock rate ratio:1.000000 Sync Count:48 PDelay Count:7

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:247] SYSTEM Clock offset:-2833129131193813 (avg:-2774105607930240) Clock rate ratio:0.999896 avg(0.999900) Sync Count:48 PDelay Count:7

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:248] QTIMER Clock offset:-1591146636253664116 (avg:-1591087612732974378) Clock rate ratio:0.999896 avg(0.999900) Sync Count:48 PDelay Count:7

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:248] Adjust clock rate ppm:-4.185667

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:377] Clock offset:0 Clock rate ratio:1.000000 Sync Count:49 PDelay Count:7

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:377] MASTER Clock offset:0 Clock rate ratio:1.000000 Sync Count:49 PDelay Count:7

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:377] SYSTEM Clock offset:-2833129131180302 (avg:-2775310169629222) Clock rate ratio:0.999896 avg(0.999900) Sync Count:49 PDelay Count:7

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2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:377] QTIMER Clock offset:-1591146636253386074 (avg:-1591088817294615429) Clock rate ratio:0.999895 avg(0.999900) Sync Count:49 PDelay Count:7

2020-05-01 08.57.53.000 daemon_cl[][info] STATUS : GPTP [08:57:53:377] Adjust clock rate ppm:-4.177938

3.1.3.4 Testing/Error Reporting

Not Applicable

4 Known Issues

Not Applicable

5 FAQs

1) **Is the below header, the client API used to connect to the daemon?**
   https://github.com/Avnu/OpenAvnu/blob/master/lib/common/avb_gptp.h

   This is not the correct file for referring the API. The basic gPTP API is gptpInit() and bool gptpGetTime(uint64_t *gptp_time_ns, uint64_t time_sys_ns)
   Please refer the helper file here
   https://source.codeaurora.org/quic/la/platform/external/OpenAVB/tree/lib/libgptp/gptp_helper.h?h=LA.AU.0.0.1_rb1.22
   This also has sample test example
   https://source.codeaurora.org/quic/la/platform/external/OpenAVB/tree/examples/libgptp_test/libgptp_test.cpp?h=LA.AU.0.0.1_rb1.22

2) **What will be the licensing agreement for this API?**

   The license should be BSD as per Licensing agreement. You can refer below link. https://source.codeaurora.org/quic/la/platform/external/OpenAVB/tree/README.rst?h=LA.AU.0.0.1_rb1.22

3) **Does Continental platform integrate the Linux PTP daemon?**

   Our Platform SW has integrated linuxptp, but this isn't recommended as Platform may not be testing/using this but recommending to use AVNU gPTP only.

6 Troubleshooting/Tips and Tricks

Not Applicable

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7 References

7.1 Abbreviations/Acronyms/Definitions

Not Applicable

7.2 Platform Documentation

TP_102G-00001: Telematics Platform Glossary

7.3 Website links

- syslog standard: https://tools.ietf.org/html/rfc5424
- syslog configuration: https://linux.die.net/man/5/syslog.conf
- Android Debug Bridge (ADB): https://developer.android.com/studio/command-line/adb
- http://www.ieee802.org/1/pages/802.1as.html

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8 Revision History

Version	Date	Description
01.00	July 01, 2020	Initial release

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